

17/07/26

Big DATA ECO SYSTEM

2. DATA INGESTION

Fundamentals of Data Science 19/04/2017
DSA0404

1. DATA SOURCES

- Social media 
- IoT Devices 
- Logs & clickstreams 
- Enterprise Applications (CRM, ERP, etc.)
- Databases 
- External Data (weather, News) 

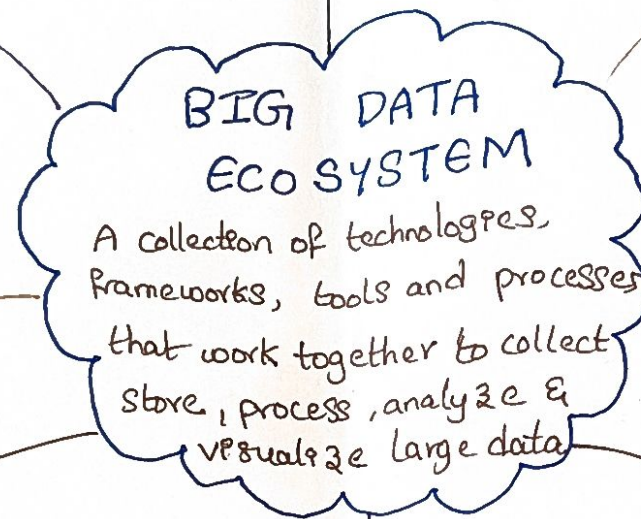
- Batch Ingestion 
- Real-time Ingestion (Streaming) 

3. STORAGE LAYER

- Data Lake (HDFS, cloud storage) 
- NoSQL databases (MongoDB, Cassandra, HBase) 

4. PROCESSING FRAMEWORK

- Batch processing (MapReduce, Spark)
- Stream processing (Spark Streaming, Flink) 



8. DATA CONSUMERS

- Business Analysts 
- Data Scientists 
- Developers 
- End users / customers 

7. MANAGEMENT & GOVERNANCE

- Data Governance 
- Security & privacy 
- Metadata Management 
- Monitoring & Logging 

6. SERVING & ACCESS

- APIs & Services 
- Applications / Dashboards
- Reports & Alerts 

5. ANALYTICS & ML

- Data Analytics (SQL, SparkSQL) 
- Machine Learning (MLlib, TensorFlow, Scikit-Learn)

KEY POINT

Big data ecosystem is all about handling the 5 V's of Big data

Right Data → Right Insight → Right Decision
Right value

COMMON TECHNOLOGIES (EXAMPLES)

hadoop Apache Spark kafka
MongoDB ++++ tableau